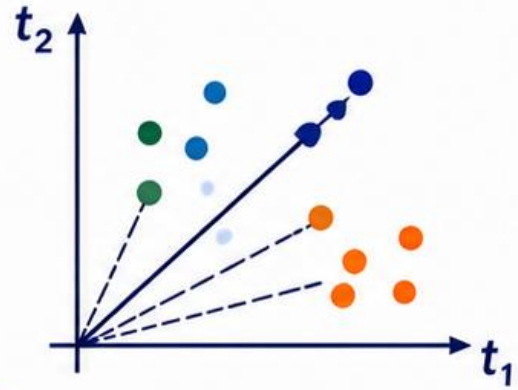
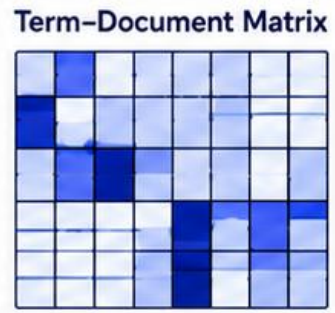
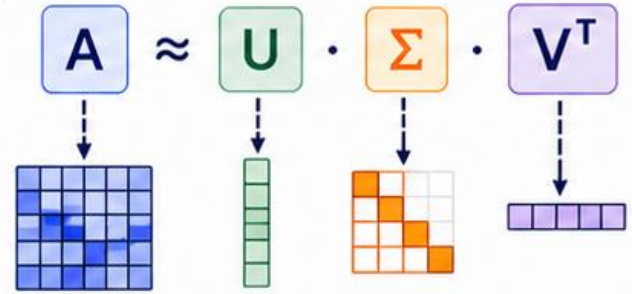
Information Retrieval /
Advanced Information Retrieval

Week 7:
Algebraic IR Models

Generalized Vector Space Model and Latent Semantic Indexing









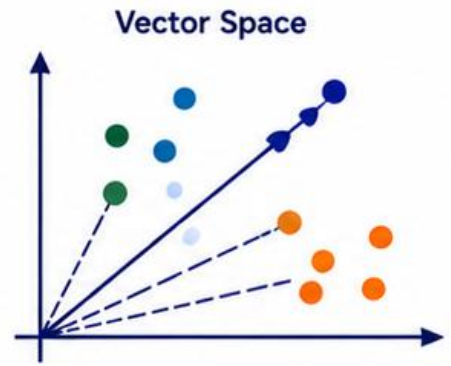
IR Focus:
Build semantic ranking and similarity-based retrieval

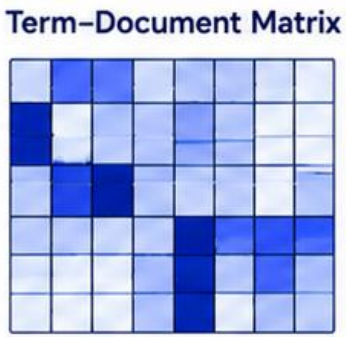
Advanced IR Focus:
Vector spaces, term correlation, matrix representation, SVD, and latent semantic dimensions

Learning Outcomes

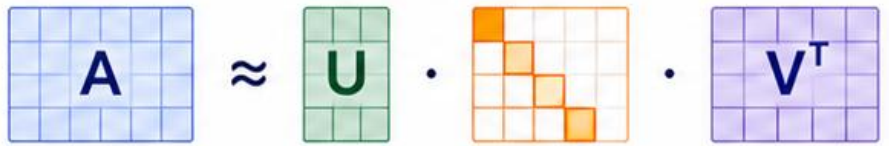
By the end of Week 7, students will be able to:

1. Explain Algebraic IR Models.
2. Understand Generalized Vector Space Model.
3. Represent documents and queries as vectors.
4. Calculate similarity using cosine similarity.
5. Explain Latent Semantic Indexing.
6. Understand the role of SVD in LSI.
7. Build semantic ranking for FYP/product systems.
8. Use algebraic IR models for research, publication, product, or funding proposals.





SVD Decomposition (LSI)



Term–Document Matrix | Term Concepts | Singular Values (Weights) | Document Concepts

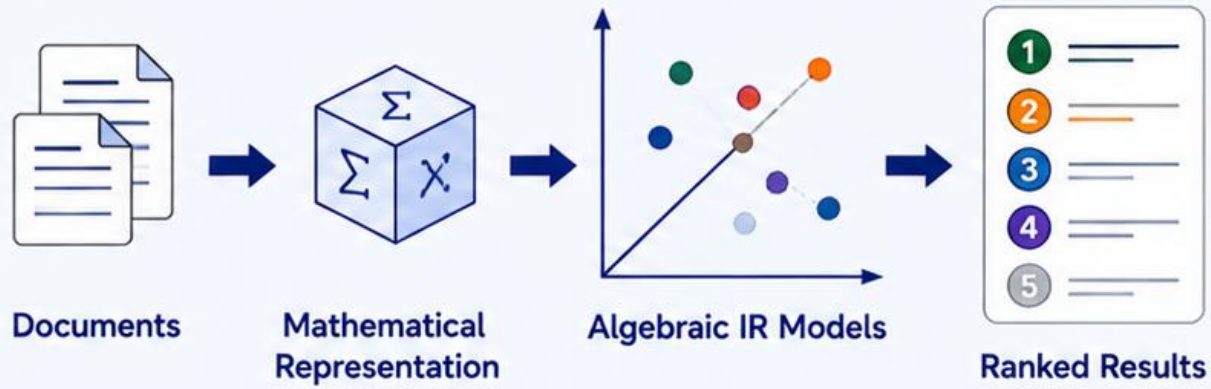
Outcome Goal:
Discover hidden (latent) meaning, measure similarity, and rank documents semantically.

What Are Algebraic IR Models?

Algebraic IR Models use **mathematical representation** to retrieve information.

They represent documents and queries as mathematical objects such as:

- Vectors
- Matrices
- Semantic dimensions
- Latent factors



They help the system go **beyond simple keyword matching**.

Why Algebraic Models Are Important

Classical models may **fail** when different words have **related meanings**.

Example:

Query:

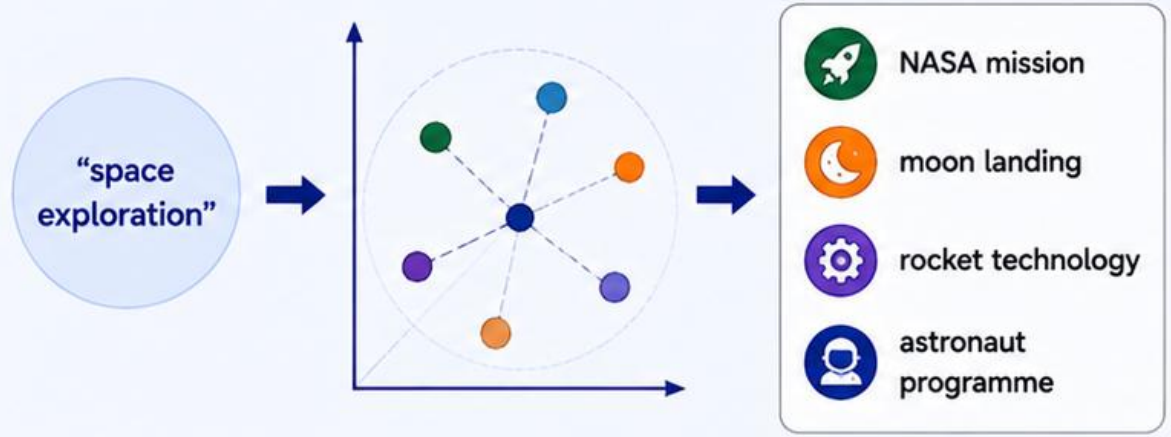
"space exploration"

Relevant documents may contain:

- NASA mission
- moon landing
- rocket technology
- astronaut programme



Algebraic models can help retrieve conceptually related documents.



Simple GVSM Example

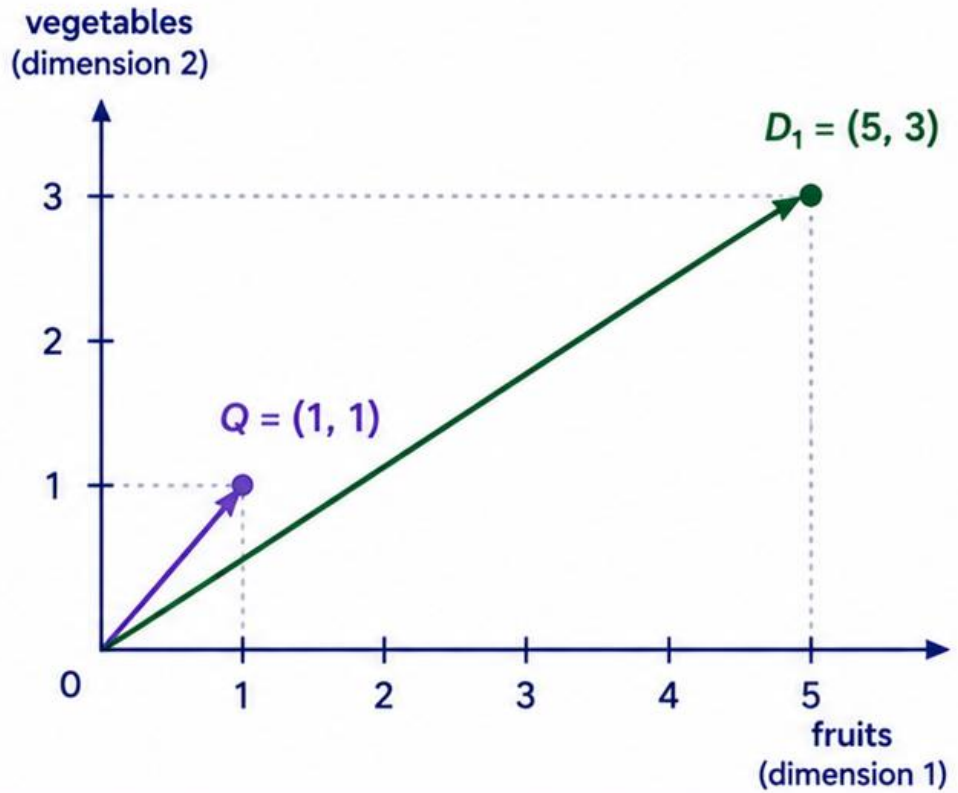
Query:
"fruits vegetables"

Document D_1:
- fruits = 5
- vegetables = 3

Document vector:
$D_1 = (5, 3)$

Query vector:
$Q = (1, 1)$

2-D Vector Space (terms as dimensions)



Interpretation

D_1 has higher values in both terms, so it is in the same direction as the query.

The system compares the direction and closeness of vectors.

Cosine Similarity

Cosine similarity measures how close the **query vector** and **document vector** are.

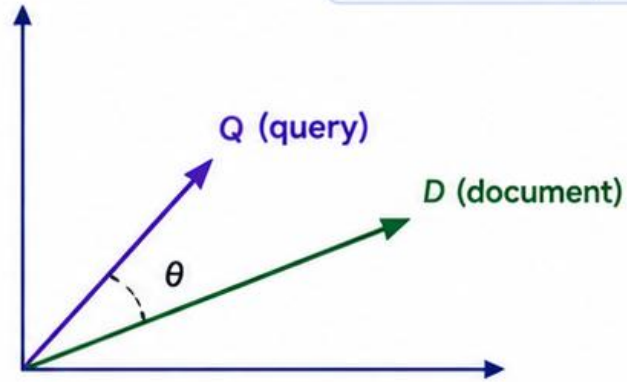
Formula:

$$\text{Cosine Similarity}(Q, D) = \frac{Q \cdot D}{||Q|| \times ||D||}$$

Where:

- $Q \cdot D$ = dot product
- $||Q||$ = length of query vector
- $||D||$ = length of document vector

Geometric Interpretation



Cosine similarity is the cosine of the angle (θ) between the two vectors.

$0 \leq \cos(\theta) \leq 1$

- 1 ⇒ same direction (high relevance)
- 0 ⇒ orthogonal (no relation)
- −1 ⇒ opposite direction

Example Calculation

Let $Q = (1, 1)$ and $D = (5, 3)$

- $Q \cdot D = (1\times5) + (1\times3) = 8$
- $||Q|| = \sqrt{1^2 + 1^2} = \sqrt{2}$
- $||D|| = \sqrt{5^2 + 3^2} = \sqrt{34}$

$$\text{Cosine Similarity}(Q, D) = \frac{8}{\sqrt{2} \times \sqrt{34}} = \frac{8}{\sqrt{68}} \approx 0.97$$

Higher cosine similarity means **higher relevance**.

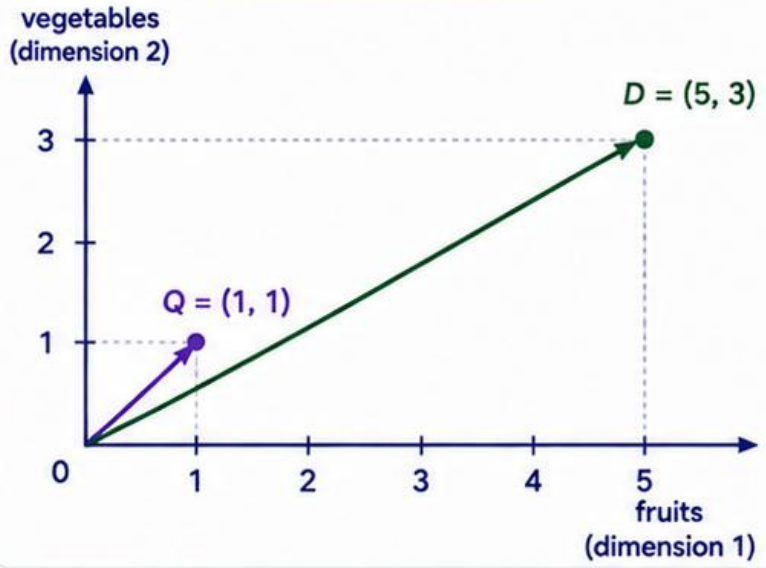
Score closer to 1 ⇒ More relevant document

Computed Example

Query:
$Q = (1, 1)$

Document:
$D = (5, 3)$

2-D Vector Space



1 Dot Product

$$Q \cdot D = (1\times5) + (1\times3) = 8$$

2 Lengths

$$||Q|| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$||D|| = \sqrt{5^2 + 3^2} = \sqrt{34}$$

3 Cosine Similarity Formula

$$\text{Cosine Similarity}(Q, D) = \frac{Q \cdot D}{||Q|| \times ||D||}$$

4 Cosine Similarity Calculation

$$\text{Cosine Similarity}(Q, D) = \frac{8}{\sqrt{2} \times \sqrt{34}}$$

$$= \frac{8}{\sqrt{2} \times \sqrt{34}}$$

$$= \frac{8}{8.24}$$

$$= \mathbf{0.97} \text{ (approximately)}$$

Result

So D is **highly relevant.**

VSM vs GVSM

GVSM extends the traditional VSM to provide more flexibility and better retrieval performance.



Feature	VSM (Vector Space Model)	GVSM (Generalized Vector Space Model)
Representation	Term vectors	Generalized term/feature vectors
Weighting	Usually TF-IDF	Flexible weighting
Similarity	Cosine similarity	Cosine or extended similarity
Flexibility	Moderate	Higher
Use	Basic ranking	Advanced ranking and research models

GVSM provides a more expressive and powerful framework for modern IR systems.

Latent Semantic Indexing

Latent Semantic Indexing goes beyond exact words.

It finds **hidden semantic patterns** in documents.

LSI may still retrieve it because the **concepts are related**.

Example

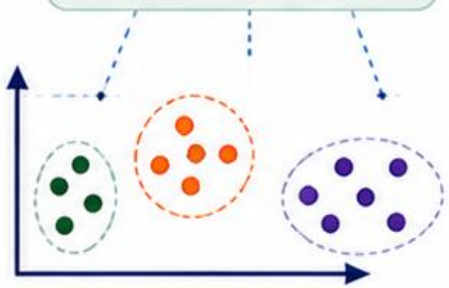
Query

"space exploration"



LSI finds latent (hidden) semantic patterns

Documents are mapped in a concept space



Document (no exact words)

May contain:

- NASA
- moon landing
- rocket
- astronaut

Even without the exact phrase "space exploration", LSI retrieves the document because the underlying concepts are **semantically related**.

Term-Document Matrix

LSI starts with a **term-document matrix**.

Rows = terms
Columns = documents
Values = term weights or presence/absence

This matrix shows which **terms** appear in which **documents**.

Example

Term-Document Matrix (A)

Terms	Documents		
	D_1	D_2	D_3
t_1	1	1	0
t_2	1	0	1
t_3	0	1	1

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

1 = term is present (or has weight)
0 = term is absent



Singular Value Decomposition

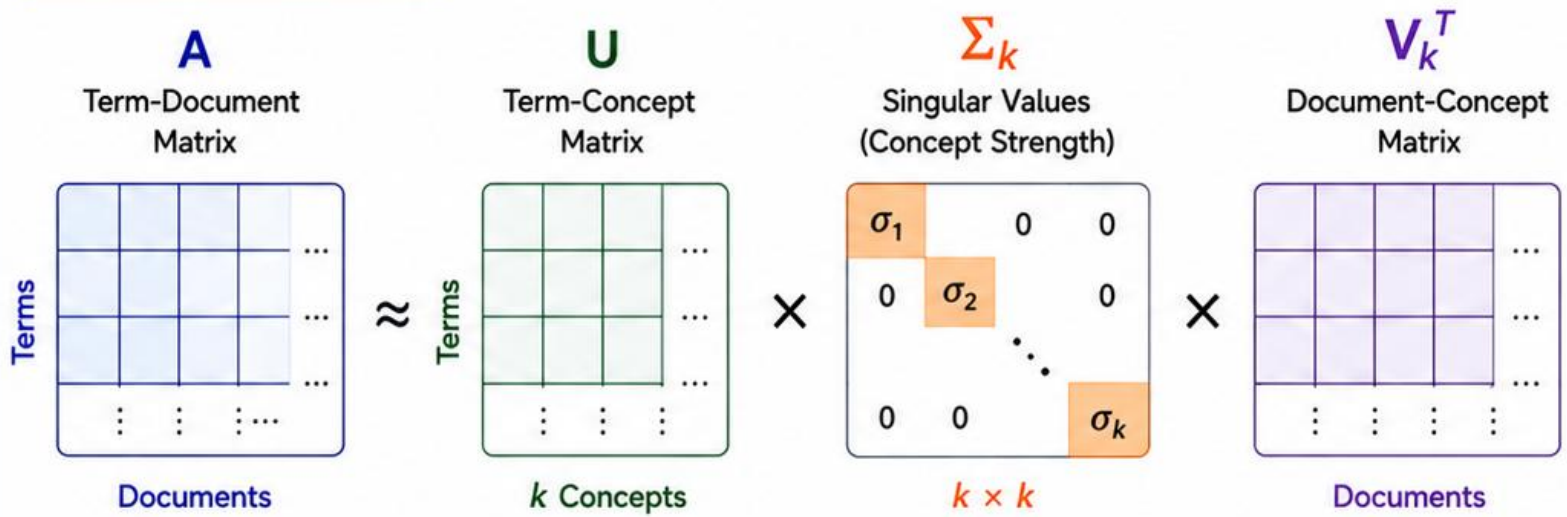
LSI applies Singular Value Decomposition:

$$A \approx U \Sigma_k V_k^T$$

Where:

- A = term-document matrix
- U = term-concept matrix
- Σ = singular values / concept strength
- V^T = document-concept matrix
- k = selected number of latent dimensions

Matrix Decomposition



We keep only the top k singular values and corresponding rows/ columns of U and V^T.

This reduces dimensionality and captures the most important patterns.

Reveals hidden semantic relationships between terms and documents.

LSI Simple Explanation



Think of LSI as finding **hidden themes.**

Example

Different Words

- space
- NASA
- rocket
- moon
- astronaut

LSI detects a hidden **concept** that connects these words.

Hidden Concept



Space Exploration

This is the hidden theme learned by LSI.

Result



LSI retrieves documents based on **meaning**, not only exact terms.

LSI understands the **meaning** behind words and retrieves the **most relevant** documents even if the exact words are different.

LSI Real-World Application

LSI helps systems understand **meaning** and retrieve **relevant content** even when words are different.



Applications

1. **Academic paper search**
   Find relevant papers even if different words are used.
2. **Movie recommendation**
   Recommend movies based on themes and meaning.
3. **Course material retrieval**
   Retrieve relevant lecture notes and study material.
4. **FYP repository search**
   Search FYP projects by concept, not just keywords.
5. **Legal document search**
   Find relevant legal cases and documents by meaning.
6. **Research article similarity**
   Identify similar articles based on semantic content.
7. **Question-answer retrieval**
   Retrieve best answers even if the question is rephrased.
8. **Semantic search module**
   Power intelligent search in modern applications.

LSI bridges the gap between how people think and how computers search.

IR Development Focus

For IR students,
Week 7 must lead to a **semantic ranking module.**



Goal: Build a **semantic ranking** search engine that understands meaning, not just exact words.

Required Features

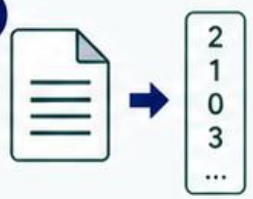
1



Document-term matrix

Create a matrix with terms as rows and documents as columns.

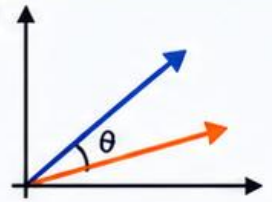
2



Vector representation

Represent documents and queries as numeric vectors (e.g., TF-IDF vectors).

3



Cosine similarity

Compute cosine similarity between query vector and document vectors.

4



Semantic similarity

Use semantic techniques to capture meaning beyond exact term matching.

5



Basic LSI idea

Apply the idea of LSI to capture hidden concepts and reduce noise.

6



Ranked search result

Return documents ranked by relevance based on semantic similarity.

7



Optional use of embeddings or matrix reduction

Enhance the system using embeddings (e.g., Word2Vec, BERT) or matrix reduction techniques.

Focus on building understanding first, then improve accuracy and performance.

Advanced IR Research/Product Focus

For Advanced IR students, Week 7 must lead to:

Move from understanding concepts to creating research and real-world solutions.



Goal: Build novel, meaningful IR solutions that contribute to research and society.

1 Mathematical explanation of GVSM

$$\vec{d}_j = \sum_{i=1}^{n} w_{ij}\ \vec{g}_i$$

Explain GVSM mathematically with generalized term vectors, weights, and similarity.

2 Matrix representation

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Represent documents, terms and concepts using matrices for computation.

3 SVD-based LSI explanation

$$A \approx \underset{m \times k}{U}\ \underset{k \times k}{\Sigma_k}\ \underset{k \times n}{V^T}$$

Explain SVD process and how LSI reduces dimensionality to uncover latent concepts.

4 Research gap in semantic retrieval



Identify open challenges and limitations in current semantic IR approaches.

5 Product concept using semantic search



Design a real-world product or system that leverages semantic search to solve a meaningful problem.

6 Published paper idea



Formulate a novel research idea suitable for publication in reputed IR/AI/CS journals or conferences.

7 Funded project proposal



Prepare a research/project proposal with objectives, methodology, impact and budget plan.

Focus on innovation, impact and contribution to knowledge.

Slide 18: Case Study Assignment

For IR Students

Build a semantic ranking module

for a selected document collection.

Minimum Requirements

1. Create document-term matrix
2. Convert query into vector
3. Calculate cosine similarity
4. Rank documents
5. Show top results
6. Explain how semantic matching improves retrieval

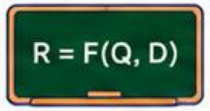
For Advanced IR Students

Prepare a research/product document

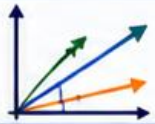
explaining GVSM and LSI mathematically with a possible publishable or fundable IR application.

- GVSM — Mathematical explanation of GVSM (Generalized Vector Space Model)
- $U \Sigma V^T$ — LSI using SVD (Singular Value Decomposition)
- Semantic ranking methodology and similarity computation
- One publishable research idea or real-world fundable application

Slide 19: Quiz

1. 

What is an Algebraic IR Model?

2. 
What is the purpose of GVSM?

3. 

What does cosine similarity measure?

4. 

What is a term-document matrix?

5. 
Why is LSI useful?

6. 

What is the role of SVD in LSI?

7. 
How can LSI help in FYP or research paper retrieval?

Slide 20: Summary

Week 7 covered:

1. 

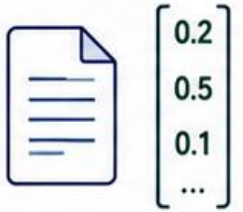
Algebraic IR Models

Formal models to represent queries, documents and relevance algebraically.

2. 

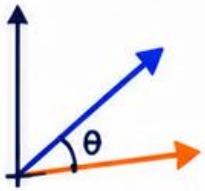
Generalized Vector Space Model

Extends the vector space model using weights and term-concept associations.

3. 

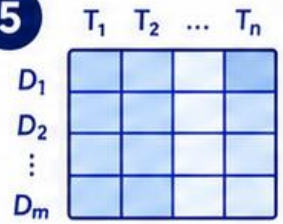
Vector Representation

Documents and queries are represented as vectors in space.

4. 

Cosine Similarity

Measures the angle between vectors to compute similarity.

5. 

Term–Document Matrix

Core matrix that shows the importance of terms in documents.

6. 

Latent Semantic Indexing (LSI)

Discovers hidden semantic structures in the data to improve retrieval.

7. 

Singular Value Decomposition (SVD)

Factorizes the term-document matrix to reduce dimensionality and noise.

8. 

Semantic Ranking

Ranks documents based on semantic similarity instead of exact matching.

9. 

FYP / Product Development

Apply IR models to build real-world systems and research contributions.